# Magic Millipedes Exhibit
# Maintenance, Care, and Troubleshooting Guide

*Organized customer-facing version*

**Quick Summary**
Below we have provided how to operate our exhibit safely and effectively. We also have included day-to-day operations to ensure successful presentations and what to do if something isn't working.

## 1. Overview

**What does the system do?**
Our exhibit is designed to be a walk-up museum box that contains a glass habitat full of millipedes. Our exhibit includes visitor indicator lights to show whether it is occupied or not, a button to begin the exhibit, a speaker to play narration about the exhibit, and a UV light to display the millipedes.

**What do you need to know?**
You don't need to know how to code. Important information includes: green means ready, red means occupied. Once someone presses the button, a two-minute presentation will run in three phases. Once the presentation is done, the light should return to green. You do not need to operate electronics during use. It is designed to be automated.

**Sequence Includes**

| Phase | Time | Experience |
|---|---|---|
| IDLE | Until button press | Green light on outside, red inside, room is available. |
| Phase 1 | 0:00–0:30 | Narration begins with countdown. Interior light stays on and red. UV is off. |
| Phase 2 | 0:30–1:30 | Interior red turns off. UV light turns on. Main narration plays. |
| Phase 3 | 1:30–2:00 | UV light turns off. Room is dark. Audio ends and system returns back to ready; exterior light is green. |

**Tech Design**
• Outside: [Green / Red status light] over door

• Inside: [Interior light over user] [button on wall with speaker module]

• Service area: [Tech box] [controller] [power] [UV light]

## 2. Normal Operation

• Check that the exterior light is green before entering.

• Check that the red light turns on on the outside once the user presses the button.

• Check that the presentation runs automatically for two minutes. Nothing is needed to be done by you here.

• The end of the presentation should produce a green exterior light.

• If the green light does not return, refer to the troubleshooting guide.

## 3. Critical Safety Warnings

| Warning | Action |
|---|---|
| Loose or damaged wires | Always check for loose or damaged wires. Live electricity can be dangerous. |
| Tech box access | Do not touch the tech box without a trained professional. |
| Liquids | Do not allow any liquids inside the exhibit or near the tech box. |
| Damaged components | If any component seems damaged or cracked, ensure that support is reached out to and close off the exhibit. |

## 4. Routine Maintenance

| Frequency | Task | Tools | Expected Result |
|---|---|---|---|
| Daily | Confirm each box shows a green light to ensure that it is in idle state. Test one full cycle per room if time permits. | N/A | Lights, button, and audio work as expected. Green returned light ensures successful completion of cycle. |
| Daily | Check daily for damaged wires or buttons. Check main power supply is plugged in. | N/A | No cracked or loose parts. Check for integrity. |
| Weekly | Clean exterior surfaces and buttons. | Cleaning supplies | Do not spray cleaner on products. |
| Weekly | Verify that UV light is | N/A | UV should only |

| | | | |
|---|---|---|---|
| | working, as this is the most important light to show light-up millipedes. | | appear during Phase 2. |
| Monthly | Open service area and check cable connections to ensure zero damage and strong connections. | Basic tools | Connections are secure and fully mounted. Only allow authorized staff to service. |

## 5. Troubleshooting Guide and FAQs

### The Room Exterior Light Never Turns Green

**Likely Cause:** Power issue, failed light, room is stuck during a cycle, or the controller did not reset.

**What to do:** Confirm it has power and ensure that green wiring is connected to the appropriate pin and the connection is secure. If it still doesn't work, report it to service.

### Button Pressed But No Start

**Likely Cause:** Button connection issue or controller fault.

**What to do:** Check that the room is green first. Press the button firmly and ensure that button pins are connected well inside the chassis. Unplug the power to the Arduino and restart if needed.

### Lights Work But No Audio

**Likely Cause:** Audio module connection issue.

**What to do:** Check speaker and audio module connections. Ask service if it is possible to see logs of whether it is detecting a player. If no player is detected, it will never run.

Most other situations can be caused by a faulty electrical outlet or faulty / damaged wire. Please reach out as soon as possible to service to see how best to troubleshoot the problem. Each situation is unique, but common fixes typically include checking the connections, doing a hard reset, or replacing a wire.

## 6. Replacement Parts / Spare Parts

| Part | Use Case | Suggested Spare | Notes |
|---|---|---|---|
| Push button | Visitor start input | 2–4 | Ensure that the button can be easily replaced and fits the mechanical guidelines. |
| Green / Red LED | Ready status | 3 | Match existing voltage and mounting type. |

| | | | |
|---|---|---|---|
| LED cap covers | Enhances exterior LED lights | 10 | In case these ever come off due to the glue not holding, replace the cap with hot glue. |
| Speaker | Audio output | 2 | Sometimes speakers become faulty for multiple reasons. Replacing the speaker with a positive and negative end can be helpful. |
| DFPlayer Mini | Audio playback | 2 | Keep one tested as a spare on hand to quickly switch out. |
| Arduino Mega | Main control center | 1 | These can be relatively easily switched out in case one encounters damage. |

## 7. Final Notes

Luckily, we will be able to provide a care and maintenance guide with all necessary parts to replace. Everything will be neatly labeled and organized so maintenance is simple whether a service provider is present or not. This will include tools and materials necessary to change parts out in the event of a faulty situation.